Association between Terrain Slope and Hip Fracture Surgery Rates Among Older Adults: A Nationwide Ecological Study in Japan

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Abstract

Hip fracture imposes a large burden on older adults in Japan, and community-level environmental correlates of hip fracture surgery rates remain incompletely understood, including terrain slope. We conducted an ecological study of 47 prefectures using the 10th National Database (NDB) Open Data: fracture surgery rates (per 100,000) from Reiwa 5 health insurance claims (April 2023 through March 2024) and habitual walking speed from Reiwa 4 Specific Health Checkup tabulations (April 2022 through March 2023); habitable-area-weighted terrain slope from national 10-m digital elevation models; aging rate and population density from the 2020 census. We fitted ordinary least squares (OLS) models with heteroskedasticity-robust (HC3) standard errors and nonparametric bootstrap resampling (*B* = 5000) as sensitivity analyses. Mean slope was 8.57 degrees (SD 3.16) and mean hip fracture surgery rate was 254.0 per 100,000 (SD 37.6). Terrain slope was positively associated with hip fracture rates in unadjusted models (β = 5.65; 95% CI, 2.50–8.79; *p* = 0.001) and in adjusted models with conventional standard errors (β = 3.49; 95% CI, 0.11–6.86; *p* = 0.043); HC3 and bootstrap intervals for the slope coefficient were borderline

コメントの追加 [山1]: 今のままだと、ちょっと難解で、統計の細かい知識がない人でも、どのような分析をしたのかがイメージできるような書き方でないと、なかなか読まれないのではないかと心配します。（本文よりも abstract の方が難解に書かれているので、本文の書き方を持ってくる方がいい） abstract を全体的にもう一度見直して、重要なメッセージがストレートに伝わるように検討してみてください。

コメントの追加 [山2]: 感度分析の話は細かいので abstract には不要ではないか。

コメントの追加 [山3]: これが何を意味するのかがわかりにくいので説明が必要と思います。

コメントの追加 [山4]: 年間１０万人当たりという意味ですか？

コメントの追加 [山5]: これが何を意味するのかがよくわかりませんでした。Discussion で使われている表現の方がわかりやすい。

コメントの追加 [山6]: これもこれだけだと何かがよくわからないと思います。何で調整したのかも書く必要があります。

at N = 47. No associations were observed for humerus or forearm fractures. Prefecture-level terrain slope showed a hip-specific signal consistent with lateral-fall mechanisms, but inferential sensitivity at small ecological N warrants cautious interpretation.

コメントの追加 **[山7]:** 抄録の最後でこのようなことは言わない方がいい。重要なことは、「傾斜が強い県に住んでいる人では股関節骨折手術が多いが、それ以外の骨折は関係ない」ことが、明確に読者に伝わることです。それ以外の細かな情報は本文で議論すればよい。

Keywords: terrain; slope; hip; fracture; ecology; Japan; epidemiology

Introduction

Hip fracture is a serious public health problem among older adults worldwide. In Japan, approximately 200,000 hip fractures occur annually, and this number is projected to increase with the ongoing demographic shift toward an aging society. [1, 2] Hip fracture leads to prolonged hospitalization, functional decline, institutionalization, and increased mortality, imposing a substantial burden on both individuals and the healthcare system.[1, 3] The acute care costs of hip fracture treatment in Japan reached approximately 1,000 billion yen ($989 million) in 2021, and this burden continues to escalate. [4, 5]

Falls are the primary cause of hip fractures, accounting for more than 90% of cases. [6] Multiple risk factors for falls have been identified, including muscle weakness, impaired gait and balance, polypharmacy, and visual impairment. [7] Among environmental factors, home hazards such as slippery floors, steps, and inadequate lighting are well recognized. [8] However, broader environmental determinants of fall risk at the community level, such as outdoor terrain characteristics, have received limited attention despite their potential population-level impact. [9]

Terrain slope is a plausible environmental determinant of fall-related fractures. Walking on inclined surfaces requires greater neuromuscular effort and alters biomechanics, increasing instability, particularly among older adults with impaired balance and reduced muscle strength. [10, 11] The control of body center of mass (CoM) during inclined walking becomes increasingly challenging with age, as margin of stability (MoS) decreases and required coefficient of friction (RCOF) changes unpredictably on uneven terrain. [12, 13] Regions with steeper terrain may expose residents to greater fall risk during daily outdoor activities, including commuting and shopping. Japan’s varied topography—ranging from flat coastal plains to steep mountain foothills—provides a natural laboratory to test this hypothesis at a nationwide scale.

Previous studies have demonstrated that environmental terrain features are associated with physical activity levels and walking speed in community-dwelling older adults. [14] Walking speed is itself a powerful predictor of fracture risk [15, 16] and has been designated the “sixth vital sign.” [17] Older adults walking on complex outdoor terrain unconsciously adopt cautious gait strategies, reducing step length and cadence, which may lead to chronic deconditioning and increased frailty. [18, 19] We therefore hypothesized that terrain slope may influence hip fracture rates through direct mechanical effects of inclined walking (direct pathway) and through its effect on walking capacity (indirect pathway).

コメントの追加 **[山8]:** なんとなく、傾斜地に住んでいる人の方が、坂を上ったり下りたりするので、足腰が鍛えられる気がしますが、どのようには考えられないですか？逆に傾斜地はしんどいので車を使う人が多いということもあるのでは？

Despite these theoretical considerations, no nationwide study has examined the association between prefectural-level terrain slope and fracture surgery rates using population-based administrative data. International ecological studies from Norway (NOREPOS) and China (CHARLS) have demonstrated significant associations between high elevation, complex terrain, and elevated hip fracture risk, [20, 21] but Japan-specific nationwide evidence has been lacking. The National Database (NDB) of Health Insurance Claims, released as open data by Japan’s Ministry of Health, Labour and Welfare, provides a unique opportunity to assess fracture surgery rates across all 47 Japanese prefectures. Combined with digital elevation model (DEM)

コメントの追加 **[山9]:** ここで県レベルというのがでてくるが、県レベルで十分なのですか？山の多い県でも実際に人々が多く住んでいるのは平野部で、山奥に住んでいる人はわずかですが、それでも代表性は担保できるのですか？limitation で少し述べられていますが、そのような欠点があっても、今回の研究を出す意義があることを discussion または conclusion で強調してください。

data from the Geospatial Information Authority of Japan, this ecological approach enables the first nationwide quantitative assessment of the terrain–fracture relationship in Japan.

In this study, we examined whether terrain slope was associated with hip fracture surgery rates at the prefectural level, independent of aging rate and walking speed, using nationwide administrative and geospatial data.

Methods

1. Study Design

We conducted a nationwide ecological study in which the Japanese prefecture (N = 47) served as the unit of analysis. Prefecture-level aggregate data were used because individual-level geospatial information is not available from NDB Open Data. The study period was fiscal year 2021 (April 2021 through March 2022).

2. Outcome: Fracture Surgery Rates

Fracture surgery rates were derived from the 10th NDB Open Data (released 2023), which contains prefecture-level counts of medical procedures claimed under the national health insurance system. We extracted claims for three fracture surgery categories: (1) hip (femur) fracture surgery, (2) humerus fracture surgery, and (3) forearm fracture surgery. Rates were calculated per 100,000 population using population estimates from the 2020 National Census.

3. Exposure: Terrain Slope

Terrain slope was calculated from a 10-m resolution digital elevation model (DEM) provided by the Geospatial Information Authority of Japan (Fundamental Geospatial Data). For each prefecture, the slope angle (degrees) was computed as the arctangent of the maximum rate of elevation change between each grid cell and its eight neighbors. We computed a habitable-area-weighted mean slope, in which each grid cell was weighted by an indicator of whether the land is classified as habitable (i.e., residential, agricultural, or commercial zones), following established methods for quantifying topographic exposure of residential populations. [10]

4. Covariate: Walking Speed

The proportion of individuals who reported walking fast (fast walking rate, %) was derived from the NDB Specific Health Checkup (Tokutei Kenko Shinsa) questionnaire administered as part of the national annual health screening program. Participants aged 40–74 years responded to a question on habitual walking speed, and we calculated the prefecture-level proportion of respondents who reported "fast" walking.

5. Additional Covariates

Aging rate (proportion of individuals aged ≥65 years, %) was obtained from the 2020 National Census. Population density (persons per km^2) was calculated as total population divided by prefecture area from the same source.

コメントの追加 **[山10]:** 老年人口割合のことですか？この書き方だと１年間にどれだけ高齢化が進んだかの指標のように見える。proportion of individuals aged ≥65 years だけでいいのでは？

6. Statistical Analysis

We summarized prefecture-level distributions using mean, SD, median, 25th and 75th percentiles, minimum, and maximum. Pearson correlation coefficients were computed between terrain slope and each outcome and covariate variable. We fit two regression models for each fracture outcome:

- **Model 1 (unadjusted)**: Simple linear regression of fracture rate on terrain slope only.
- **Model 2 (multivariable)**: Multiple linear regression of fracture rate on terrain slope, aging rate, fast walking rate, and population density. In the primary causal interpretation, aging rate and fast walking rate were treated as confounder-adjustment variables rather than formal mediators.

コメントの追加 [治齋11]: （佐藤先生コメント①）完全な媒介因子として考えることは難しいので、交絡因子として処理する旨を記載しました。

Model performance was assessed using the coefficient of determination (R^2), adjusted R^2, and the F-statistic. Regression coefficients (β) and 95% confidence intervals (CIs) are reported. Statistical significance was defined as $p < 0.05$ (two-sided). All analyses were performed using Python (version 3.14.2) with the statsmodels library (version 0.14.6). Linear regression models were estimated by ordinary least squares (OLS) using the statsmodels formula interface.

For the primary hip-fracture outcome (Model 2), we assessed linear model assumptions graphically using residuals-versus-fitted and normal quantile–quantile (Q–Q) plots of OLS residuals, and we reported the Shapiro–Wilk test on residuals for descriptive completeness (small-N tests are interpreted cautiously). Because prefecture-level ecological analyses can exhibit heteroskedasticity and leverage from skewed covariates (notably population density), we conducted sensitivity analyses: (1) heteroskedasticity-robust standard errors (HC3) for two-sided p values and CIs, and (2) a nonparametric bootstrap with B = 5,000 resamples of prefectures (with replacement) to obtain a percentile 95% CI for the terrain slope coefficient. Given N = 47 with multiple covariates, we also checked coefficient stability and multicollinearity diagnostics (including model condition number) and interpreted estimates cautiously.

コメントの追加 [治齋12]: （佐藤先生コメント②）N=47 での係数安定性・多重共線性診断を確認し、慎重解釈する旨を追記しました。

As an additional sensitivity analysis for age-structure confounding, we performed indirect standardization for the femur-surgery outcome. We combined NDB age-stratified surgery tables with 2020 census 5-year age-group population counts from e-Stat. Because disclosure-control masking (`"-"`) was present in age-stratified surgery cells, we prespecified imputation scenarios (`"-" = 0`, `5`, or `9`) and repeated the SIR/ISR workflow for each scenario. For each scenario, we calculated national age-specific surgery rates, expected prefectural counts, SIR, and ISR (SIR × national crude rate). We then regressed ISR on terrain slope with and without covariate adjustment. These analyses were treated as sensitivity analyses only (not primary inference), given potential secondary masking.

コメントの追加 **[治齋13]:** （寶澤先生コメント）年齢階級別手術表データ、2020 年国勢調査の人口データを用いた間接法 SIR/ISR の感度分析手順（`0/5/9`シナリオ）を追記しました。

All manuscript figures were generated programmatically from study data (not by image-generative AI tools) using reproducible Python workflows: Python 3.14.2 with pandas 2.3.3, numpy 2.3.5, matplotlib 3.10.8, seaborn 0.13.2, scipy 1.16.3, geopandas 1.1.2, and japanize matplotlib 1.1.3. Figure 1 was generated via linear-fit visualization from the prefecture-level analytical dataset, Figure 2 from the corresponding correlation matrix, and Figures 3-5 by joining prefecture-level indicators to national prefectural boundary geodata and rendering choropleth maps.

コメントの追加 **[山14]:** これは不要では？

This study used publicly available aggregate data; individual informed consent was not required, and institutional ethics review was not applicable in accordance with Japanese ethical guidelines for epidemiological research.

Results

1. Descriptive Statistics

Across the 47 Japanese prefectures, the mean habitable-area-weighted terrain slope was 8.57 ± 3.16 degrees (median 8.36; interquartile range [IQR] 6.56–10.95; range 1.81–15.43 degrees) (Table 1). The mean hip fracture surgery rate was 254.0 ± 37.6 per 100,000 population (median 253.9; IQR 228.6–286.3; range 167.6–329.6), and the mean total fracture surgery rate was 352.8 ± 54.9 per 100,000 (median 356.1; IQR 306.4–397.1). The mean aging rate was 30.7 ± 3.1% (median 31.3%; IQR 29.3–32.9%), and the mean proportion with fast walking speed was 48.6 ± 4.3% (median 49.2%; IQR 46.2–51.2%). Population density was right-skewed (mean 657 persons/km^2 versus median 265; IQR 173–469).

2. Correlation Analysis

Terrain slope showed a moderate positive correlation with hip fracture surgery rate (r = 0.475, $p < 0.01$) and with total fracture surgery rate (r = 0.394, $p < 0.01$) (Table 2). Figure 1 illustrates the bivariate association between terrain slope and hip fracture surgery rate; Figure 2 displays the correlation matrix. Slope was also positively correlated with aging rate (r = 0.505, $p < 0.01$). In contrast, slope showed a weak negative correlation with fast walking rate (r = −0.264). Slope showed weak and non-significant correlations with humerus fracture rate (r = 0.138) and forearm fracture rate (r = 0.177).

コメントの追加 [山15]: rate という単語は、疫学では、単位時間当たりを含む概念なので、単に割合を意味するなら proprotion の方がいい。ほかも同様です。

コメントの追加 [山16]: これは Table2 と何が違いますか？どちらかだけでいいのでは？

3. Regression Analysis: Hip Fracture

In the unadjusted model (Model 1), terrain slope was significantly and positively associated with hip fracture surgery rate (β = 5.65; 95% CI, 2.50–8.79; R^2 = 0.225; p =

0.001). After adjustment for aging rate, fast walking rate, and population density (Model 2), the slope coefficient was 3.49 (95% CI, 0.11–6.86; R^2 = 0.385; *p* = 0.043) using conventional (model-based) standard errors (Table 3). Aging rate was also associated with hip fracture rate (β = 5.53; 95% CI, 1.32–9.74; *p* = 0.011). Fast walking rate was positively associated with hip fracture rate under HC3 robust inference (β = 2.02; 95% CI, 0.29–3.74; *p* = 0.022) but not under conventional standard errors (β = 2.02; 95% CI, −0.34–4.37; *p* = 0.091). Population density was not significantly associated with hip fracture rate (β = −0.001; *p* = 0.818).

Residual diagnostics for Model 2 did not indicate gross deviations from linearity in residuals versus fitted values. The Shapiro–Wilk test on OLS residuals was *W* = 0.957 (*p* = 0.079). Sensitivity analyses, however, indicated inferential fragility for the primary slope coefficient at N = 47: heteroskedasticity-robust (HC3) 95% CIs were −0.19 to 7.17 (*p* = 0.063), and a nonparametric bootstrap 95% CI (prefecture resampling, *B* = 5,000) was −0.03 to 7.09 (both intervals include null).

In masking-imputation indirect-standardization sensitivity analyses, the slope coefficient remained positive across all scenarios. In multivariable models (ISR ~ slope + aging rate + fast walking rate + population density), β was 5.71 (95% CI, 0.54–10.88; *p* = 0.031) for `"-" = 0`, 3.61 (95% CI, 0.30–6.92; *p* = 0.033) for `"-" = 5`, and 3.64 (95% CI, 0.28–7.00; *p* = 0.035) for `"-" = 9`. Thus, the direction of association was stable, whereas the effect size varied across masking assumptions.

4. Regression Analysis: Total, Humerus, and Forearm Fractures

In the unadjusted model, slope was significantly associated with total fracture surgery rate (β = 6.85; 95% CI, 2.06–11.64; R^2 = 0.156; *p* = 0.006). In the multivariable model, however, slope did not reach statistical significance for total fracture rate (β = 4.47; *p* = 0.099), although aging rate remained a significant predictor (β = 7.19; *p* = 0.035; R^2 = 0.276). Neither humerus fracture rate (Model 1: *p* = 0.355; Model 2: R^2 =

コメントの追加 [山17]: これは必要ですか？

コメントの追加 [治齋18]: （寶澤先生コメント）Results に、各シナリオでの ISR 回帰係数（β, 95%CI, *p*）を明記し、符号は安定・効果量は変動する旨を明示しました。

コメントの追加 [山19R18]: "-"が何を意味するのかがわからないので、「秘匿になっているセルを 0 と仮定したとき」など、読者にわかりやすく書く方がいいと思います。

0.107) nor forearm fracture rate (Model 1: p = 0.233; Model 2: R^2 = 0.062) was significantly associated with terrain slope in either model.

Discussion

1. Main Findings

This ecological study showed~~demonstrated~~ a positive association between prefectural-level terrain slope and hip fracture surgery rates after adjustment for aging rate, walking speed, and population density (β = 3.49 per one-degree increase; conventional 95% CI, 0.11–6.86; p = 0.043), while acknowledging that heteroskedasticity-robust and bootstrap intervals for the slope coefficient were compatible with no association at α = 0.05. Each additional degree of slope corresponded to approximately 3.5 additional hip fracture surgeries per 100,000 population under the primary OLS parameterization. The adjusted association was specific to hip fractures and was not observed for humerus or forearm fractures. These findings suggest that, at the population level, steeper habitable terrain may track hip-specific surgical burden beyond compositional aging and self-reported walking pace.

コメントの追加 [治齋20]: （佐藤先生コメント⑩）demonstrated → showed と、弱い表現に修正しました。

コメントの追加 [山21]: abstract もこのような書き方の方が理解されやすいと思います。

コメントの追加 [山22]: ここも同様に、abstract をこのように書く方がわかりやすいと思います。

コメントの追加 [山23]: そのような県に住んでいること、との関連という方がこの研究の特徴を表せるように思います。

2. Biological Plausibility

The ~~specificity of the~~ slope–fracture association observed for~~to~~ the hip may be~~is~~ biologically plausible and may be partly supported by ~~extensive~~ biomechanical evidence. Hip fractures typically result from lateral (sideways) falls, [22, 23] which are common~~the predominant~~ fall mechanisms on inclined surfaces. Real-world video-capture studies of falls in long-term care facilities have demonstrated that lateral landing configuration increases hip fracture risk by 5.5 to 6-fold compared to forward or backward falls, and direct impact to the greater trochanter elevates fracture risk by up to 30-fold. [23, 24] The impact forces generated during lateral falls typically reach 2,000 to 4,000 N, exceeding the physiological tolerance of osteoporotic femoral necks. [25] Finite element method (FEM) simulations have further suggested~~confirmed~~ that posterolateral and lateral impacts generate maximum tensile and compressive stresses at the superior and inferior aspects of the femoral neck, potentially increasing~~maximizing~~ fracture vulnerability. [26]

コメントの追加 [治齋24]: （島袋先生コメント）考察の biological plausability の次の一文について、トーンを緩めた表現に修正いたしました。

When walking downhill or across a slope, the body's center of mass (CoM) shifts laterally, and the control of lateral stability becomes increasingly difficult. [27, 28] The human gait pattern is biomechanically analogous to an inverted pendulum, with CoM tracing a bow-tie-shaped trajectory in three-dimensional space. [28] On sloped terrain, maintaining CoM within the base of support (BoS) requires precise step-by-step adjustments of step width and foot placement. [29] For older adults with age-related declines in hip abductor strength and ankle dorsiflexion, the margin of stability (MoS) during inclined walking is dramatically reduced. [11, 13] Slips or minor perturbations that could be corrected via cross-over steps on flat surfaces become unarrested lateral falls on slopes due to delayed postural reflexes and environmental constraints. [12, 28]

In contrast, wrist (forearm) fractures are more commonly caused by protective extension reflexes during forward falls on flat surfaces. [30] When falling forward, individuals instinctively extend their arms to protect the head and torso, and kinematic studies show that elbow flexion and wrist dorsiflexion absorb fall energy, dramatically reducing pelvic impact velocity. [30] Such forward falls typically occur from tripping on minor obstacles or steps indoors, rather than from rapid lateral displacement on steep slopes. [31] Similarly, humerus fractures in older adults often result from low-energy obliquely forward falls on level ground and occur in relatively younger, more ambulatory individuals who retain arm-protective reflexes. [32] The fracture-site specificity observed in our data—significant association with hip fractures but not with humerus or forearm fractures—~~thus~~ provides internal evidence that is consistent with ~~the~~ a biomechanical paradigm in which~~that~~ terrain slope may specifically amplify~~ies~~ lateral fall risk.

コメントの追加 **[治齋25]:** （佐藤先生コメント⑩）
Discussion 内の断定表現を修正しました。

3. Comparison with Previous Studies

Our findings align with and extend international ecological studies examining geospatial determinants of hip fracture risk. The Norwegian Epidemiologic Osteoporosis Studies (NOREPOS), conducted in one of the world's highest hip fracture incidence countries, used GIS-linked nationwide hospital records to demonstrate that residents of inland, high-elevation, topographically complex areas experienced significantly higher hip fracture incidence compared to those in flat coastal regions. [20] The NOREPOS investigators further identified that cold ambient temperature and icy terrain in mountainous areas amplified this geographic risk, [33] a finding with direct relevance to Japan's snowy mountainous prefectures (e.g., Niigata, Nagano, Yamagata).

In Asia, a recent retrospective cohort analysis using data from the China Health and Retirement Longitudinal Study (CHARLS) demonstrated that exposure to high altitude and steep terrain was associated with a 1.65-fold elevated hazard ratio for hip fracture, with particularly pronounced effects among overweight individuals and women. [21] These findings reinforce the generalizability of our results across diverse geographic and ethnic contexts.

Previous ecological studies have also examined the relationship between outdoor environment and fall-related injuries. Studies in multiple countries found that neighborhood green space, walkability, and environmental safety were associated with fall-related hospital admissions and fracture risk. [34] In Japan, Nakamura et al. reported that frequent outdoor activity predicts physical function in rural community-dwelling older adults. [14] However, to our knowledge, no prior study has used nationwide DEM data linked to administrative fracture data to quantify the dose-response association between terrain slope and hip fracture rates at the population level in Japan. Our findings add~~study provides the first~~ nationwide quantitative evidence that each one-degree increase in terrain slope is associated with approximately 3.5 additional hip fracture surgeries per 100,000 population, independent of aging rate in the primary model. This quantitative estimate ~~is novel and~~ may contribute to the global evidence base for terrain as a modifiable environmental correlate of fracture burden~~risk factor~~.

コメントの追加 [治齋26]: （佐藤先生コメント⑩）Discussion 内の断定表現を修正しました。

4. Role of Aging Rate

Aging rate was the strongest predictor of hip fracture rate in the multivariable model (β = 5.53, p = 0.011). Prefectures with steeper terrain also tended to have higher aging rates (r = 0.505), reflecting the well-documented trend of rural aging in Japan, where mountainous prefectures experience greater outmigration of younger populations. [35, 36, 37] Older adults in these high-slope prefectures thus face a double burden: the physical challenge of navigating harsh terrain on a daily basis (environmental hazard) and the social isolation and lack of medical access associated with rural depopulation (social determinants of health). [38, 39]

The partial association of slope with hip fracture after adjustment for aging rate suggests that terrain may constitute a risk correlate~~factor~~ over and above demographic composition, subject to the inferential sensitivity noted above. This finding has practical implications: even within the context of Japan's nationwide aging, geographic topography may represent~~s~~ an additional and potentially modifiable layer of fracture risk. Terrain slope may~~is~~ not be merely a proxy for aging and may represent~~, but~~ an additional~~independent~~ environmental correlate~~hazard~~ that could contribute to functional decline~~chronically erodes physical function~~ among rural-dwelling older adults. Given public-data masking in age-stratified surgery tables, we interpret age-structure robustness primarily from the consistency of adjusted ecological models and from imputation-based sensitivity analyses, while acknowledging that full indirect standardization without imputation is not feasible with the current open dataset.

コメントの追加 [治齋27]: （佐藤先生コメント⑩）Discussion 内の断定表現を修正しました。

コメントの追加 [治齋28]: （賓澤先生コメント）Discussion に、NDB オープンデータのマスキング制約下での解釈方針（頑健性確認として扱う旨）を追記しました。

5. Walking Speed as a Mediator

At the individual level, slower walking speed is a well-established marker of fracture risk, sarcopenia, frailty, and mortality (often termed the "sixth vital sign"). [15, 16, 17] Prospective cohort studies in community-dwelling older adults have demonstrated that walking speeds below 0.8 m/s or 0.69 m/s significantly elevate hip fracture incidence. [16] Walking speed measured in clinical settings (flat 10-meter walk tests) may not accurately reflect real-world walking ability on complex outdoor terrain. [18]

In our ecological data, terrain slope was negatively correlated with prefecture-level fast walking rate (r = −0.264), consistent with residents of steeper prefectures reporting slower habitual walking on average. When confronted with uneven or sloped terrain, older adults unconsciously adopt cautious gait strategies, reducing step length and cadence to enhance stability, which results in substantially slower walking speeds. [11, 18, 19] Older adults residing in high-slope regions are exposed to such challenging terrain daily during shopping and outdoor activities, leading to chronic fear of falling, reduced outdoor activity, and progressive deconditioning (disuse muscle atrophy). [40, 41]

コメントの追加 [山29]: 車の影響が考察されていないのが気になります。地方で傾斜地で高齢者ならほぼ車を使う人ばかりと思います。

However, the multivariable OLS coefficient for fast walking rate was **positive** (β = 2.02 per 1 percentage-point increase in fast walking), which is difficult to interpret as a causal protective effect at prefecture resolution and is not the individual-level association described in cohort studies. Inference for this coefficient was sensitive to standard error choice (conventional $p = 0.091$ versus HC3 $p = 0.022$). Plausible drivers include ecological bias, aggregation of a 40–74-year screening metric to represent regions where hip fractures concentrate among the oldest-old, and confounding by unmeasured healthcare and reporting behaviors. Because only the 40–74-year questionnaire aggregate is publicly available, we treated this variable as a contextual covariate and not as a direct proxy for hip fracture risk in the oldest age groups. We therefore did not treat walking speed as a statistically supported mediator of the slope–hip fracture association. Future studies using individual-level data (e.g., objectively measured gait speed, accelerometry) with formal causal mediation analysis are warranted to clarify whether terrain slope influences hip fracture risk partly through its chronic effect on habitual walking capacity.

コメントの追加 **[治齋30]:** （佐藤先生コメント⑧）40–74 歳質問票集計における公開であるため、歩行速度変数は contextual covariate として扱う旨を記載しました。

6. Policy Implications

Our findings may have concrete policy implications in the context of Japan's national health promotion strategies and the ongoing escalation of hip fracture healthcare costs. The NDB trends from 2012 to 2023 demonstrate that hip fracture incidence, which had plateaued temporarily, resumed an upward trajectory after 2021, likely reflecting COVID-19-related outdoor activity restrictions and accelerated frailty. [4, 42, 43] Acute care costs for hip fracture treatment reached approximately 1,000 billion yen in 2021, [5] prompting the Japanese government to introduce a new reimbursement incentive in April 2022 for early surgery within 48 hours for patients aged ≥75 years. [44]

First, prefectures with steeper terrain should be prioritized in national fall prevention programs. Japan's Ministry of Health, Labour and Welfare has promoted the "Active Guide" (Plus-Ten campaign: "add 10 minutes of physical activity daily") under the Health Japan 21 (second edition) framework. [45, 46] However, research has shown that physical activity behavior is strongly dependent on neighborhood environmental features (sidewalk availability, safety, topography), not individual motivation alone. [45] Recommending “10 additional minutes of outdoor walking” in mountainous prefectures with mean slopes exceeding 10 degrees (e.g., regions in Shikoku, Chugoku, inland Kyushu) may inadvertently increase lateral fall and hip fracture risk without concurrent environmental modifications. High-slope regions require environment-specific adaptations of the Active Guide, including balance and strength training programs, anti-slip footwear distribution, and handrail installation subsidies for outdoor pathways. [47]

Second, housing modification subsidies targeting high-slope prefectures—such as installation of handrails on outdoor pathways and anti-slip surface materials—could reduce the environmental contribution to fall risk. Third, urban planning guidelines for mountainous communities should explicitly incorporate slope gradient as a factor in pedestrian infrastructure design, particularly in areas with high proportions of older residents. Given the irreversible demographic aging in rural Japan, these upstream environmental interventions represent cost-effective strategies to mitigate the downstream burden of fragility fractures.

コメントの追加 **[山31]:** 根拠がなければ、やや言い過ぎのように思われます。

7. Strengths

This study has several strengths. We used nationally representative administrative data covering all 47 Japanese prefectures, ensuring that results were not subject to selection bias affecting individual-level studies. The use of DEM-derived weighted slope measures provides an objective, reproducible quantification of terrain exposure. The fracture outcomes were ascertained from insurance claims, which are routinely recorded with high administrative accuracy. The fracture-site specificity observed (hip but not forearm or humerus) provides internal evidence consistent with the proposed biomechanical mechanism.

8. Limitations

Several limitations should be considered. First, as an ecological study, our findings are subject to the ecological fallacy: associations observed at the prefecture level may not reflect individual-level relationships. [48] We cannot establish whether individuals living in steeper terrain within a prefecture have higher fracture risk than those in flatter areas of the same prefecture. Individual-level studies with residential geocoding and objective measures of daily terrain exposure (e.g., GPS tracking, accelerometry) are necessary to confirm our prefecture-level findings.

Second, our outcome reflects surgery rates rather than true fracture incidence. Although hip fractures are usually treated surgically, prefectural variation in treatment decisions, referral patterns, and access to operative care may influence observed rates independently of fracture occurrence. Third, the cross-sectional design precludes causal inference. Although reverse causation is unlikely for a fixed environmental exposure such as terrain slope, residual confounding by unmeasured variables cannot be excluded. Fourth~~Third~~, important unmeasured confounders—including individual-level socioeconomic status, healthcare access (e.g., prefectural orthopedic surgeon density, distance to emergency hospitals), nutritional status (calcium and vitamin D intake), bone mineral density (BMD), climate (snowfall and icing), and medication use (bisphosphonates, vitamin D supplementation)—may account for part of the observed association. Future studies incorporating NDB prescription data and BMD screening results at finer geographic resolution would strengthen causal inference.

コメントの追加 [治齋32]: （佐藤先生コメント③）Limitations に「surgery rates rather than true fracture incidence」と記載を加えました。また、治療方針・アクセス差の影響がある可能性について記載しました。

コメントの追加 [治齋33]: （佐藤先生コメント④）気候（積雪・凍結）の項目について、Limitations に追加しました。

Fifth, age-standardization sensitivity results relied on imputation of masked (`"-"`) cells in publicly released age-stratified surgery tables. Although directionality was robust across `0/5/9` scenarios, effect size depended on masking assumptions, and secondary masking rules imply that some masked cells may not correspond strictly to the nominal `<10` range. Therefore, these SIR/ISR analyses should be interpreted as robustness checks rather than definitive age-standardized estimates. Sixth~~Fourth~~, the walking speed variable was based on self-reported habitual walking speed in the Specific Health Checkup questionnaire, administered to individuals aged 40–74 years, because publicly available prefecture-level data were limited to individuals aged 40–74 years in the current NDB Open Data. This variable ~~which~~ may not accurately represent the functional capacity of the oldest-old (≥75 years) at highest fracture risk, and its direct relevance to hip fracture occurrence is likely limited at this ecological resolution. This self-reported measure may be subject to non-differential measurement error, biasing our estimates toward the null. Seventh~~Fifth~~, the DEM-based slope calculation reflects the topographic exposure of the prefecture as a whole and does not capture micro-scale terrain variations (e.g., steep residential pathways, stairways) encountered during daily activities by individual residents. High-resolution spatial analysis at the municipal or neighborhood level would provide more granular insights into the terrain–fracture relationship.

Eighth~~Sixth~~, with only 47 prefectures, regression inference can be sensitive to heteroskedasticity and leverage from skewed covariates (especially population density). Our sensitivity analyses (HC3 robust standard errors and nonparametric bootstrap resampling of prefectures) indicated that the conventional $p = 0.043$ result for the primary slope coefficient in Model 2 was borderline and compatible with no association under alternative variance assumptions. Ninth~~Finally~~, sex- and gender-stratified outcome tables were not available in the prefecture-level NDB Open Data extracts used here; findings therefore describe aggregate populations and should be interpreted with this limitation in mind.

コメントの追加 [治齋34]: （寶澤先生コメント）Limitations に、補完依存性と二次マスキングの可能性を明記しました。

コメントの追加 [治齋35]: （佐藤先生コメント⑧）歩行速度指標が 40–74 歳の公開質問票集計に基づくことを明記し、最高齢群の骨折リスクを直接代表しないという限界が伝わるように修正しました。

Conclusions

Under primary ordinary least squares estimation, prefectural terrain slope was~~remained~~ positively associated with hip fracture surgery rates after adjustment for aging rate, walking speed, and population density (β ≈ 3.5 additional surgeries per 100,000 per one-degree increase; conventional *p* = 0.043). The adjusted signal was specific to hip fractures, with no significant relationship observed for humerus or forearm fractures. Future studies using individual-level data with residential geocoding, objective gait measurements, formal causal mediation analysis, and high-resolution spatial modeling are warranted to delineate the precise causal pathways—direct biomechanical effects versus chronic functional decline—underlying the terrain–fracture association observed at the population level.

コメントの追加 **[治齋36]:** （佐藤先生コメント⑩）
remained → was と、弱い表現に修正しました。

Data availability

Aggregate data used in this study are publicly available from the Japanese National Database Open Data and official statistics cited in the Methods. Under Ministry of Health, Labour and Welfare distribution rules, individual-level claims and identifiable records cannot be shared. Prefecture-level values supporting this analysis can be reproduced from those public sources using the procedures described. We encourage deposition of analysis code and derived prefecture-level tables in a public repository (e.g., Zenodo) with a DOI and citation in this article, in line with the journal's research data guidance (Option B). Author contributions will be reported using the CRediT taxonomy on the separate title page required for double-anonymized submission.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation and writing of this work, the authors used AI-assisted tools to support manuscript drafting and statistical analysis scripting. Cursor 2.0 (Anysphere) and Google Antigravity (Google) were used for AI-assisted writing and Python code development. Large language models used through these platforms included Claude Sonnet 4.6 and Claude Opus 4.6 (Anthropic) and Gemini 3 Pro (Google). These tools were used only for text drafting and code generation; no generative AI or AI-assisted tools were used to create, alter, or otherwise process any figures, images, or artwork in this manuscript. The authors reviewed and edited all AI-assisted outputs and were responsible for the study design, selection of statistical methods, interpretation of findings, conclusions, and final reference list. The authors take full responsibility for the integrity and accuracy of the final content. AI was not listed as an author.

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Tables and Figures

Table 1. Descriptive Statistics of Prefecture-Level Variables (N = 47)

Variable	Unit	N	Mean	SD	Median	Q1	Q3	Min	Max
Outcome variables									
Total fracture surgery rate	per 100,000	47	352.75	54.85	356.11	306.44	397.13	249.22	446.68
Hip (femur) fracture surgery rate	per 100,000	47	253.99	37.57	253.87	228.57	286.32	167.61	329.62
Humerus fracture surgery rate	per 100,000	47	35.59	7.52	35.56	30.04	41.72	22.39	50.45
Forearm fracture surgery rate	per 100,000	47	63.18	15.59	60.84	51.68	76.04	33.49	106.33
Exposure variable									
Habitable-area-weighted slope	degrees	47	8.57	3.16	8.36	6.56	10.95	1.81	15.43
Simple mean slope	degrees	47	8.28	2.90	8.18	6.37	10.24	1.88	14.22
Covariate variables									

Variable	Unit	N	Mean	SD	Median	Q1	Q3	Min	Max
Aging rate (≥65 years)	%	47	30.74	3.10	31.26	29.28	32.88	22.61	36.45
Fast walking rate	%	47	48.62	4.34	49.23	46.23	51.22	29.79	57.30
Population density	persons/km^2	47	657	1,223	265.44	173.20	468.68	62.63	6,402.64

Note. Q1 and Q3 denote 25th and 75th percentiles (prefecture-level observations). Exploratory Shapiro–Wilk tests rejected normality for fast walking rate and population density (both $p < 0.001$); other listed variables were not rejected at $\alpha = 0.05$ (aging rate $p = 0.090$). Marginal normality is not required for ordinary least squares; median and quartiles are presented to describe skewed distributions alongside mean and SD.

Table 2. Pearson Correlation Matrix of Study Variables (N = 47)

Variable	Hip Fracture	Total Fracture	Humerus Fracture	Forearm Fracture	Terrain Slope	Aging Rate	Fast Walking Rate	Pop Density
Hip Fracture	1.000							
Total Fracture	0.962***	1.000						
Humerus Fracture	0.697***	0.801***	1.000					
Forearm Fracture	0.640***	0.813***	0.656***	1.000				
Terrain Slope	0.475***	0.394**	0.138	0.177	1.000			
Aging Rate	0.533***	0.430**	0.152	0.156	0.505***	1.000		
Fast Walking Rate	−0.044	0.011	0.177	0.060	−0.264	−0.406**	1.000	
Population Density	−0.336*	−0.238	0.006	−0.032	−0.345*	−0.628***	0.381**	1.000

p < 0.05 (), p* < 0.01 (), **p < 0.001 (***).

Table 3. Linear Regression Analysis: Association between Terrain Slope and Hip Fracture Surgery Rate (N = 47)

	Model 1 (Unadjusted)		**Model 2** (Multivariable)	
Variable	**β (95% CI)**	***p***	**β (95% CI)**	***p***
Terrain slope (per 1 degree)	**5.65 (2.50, 8.79)**	**0.001**	**3.49 (0.11, 6.86)**	**0.043**
Aging rate (per 1%)	—	—	**5.53 (1.32, 9.74)**	**0.011**
Fast walking rate (per 1%)	—	—	2.02 (−0.34, 4.37)	0.091
Population density (per 1 person/km^2)	—	—	−0.001 (−0.01, 0.01)	0.818
R^2	0.225		0.385	
Adjusted R^2	0.208		0.326	
F-statistic (*p*)	13.23 (0.001)		6.56 (0.0003)	

Model 1: simple linear regression of hip fracture rate on terrain slope alone. Model 2: multivariable linear regression adjusting for aging rate, fast walking rate, and population density. β, unstandardized regression coefficient; CI, confidence interval. **Bold** indicates $p < 0.05$ (conventional model-based standard errors).

Sensitivity (Model 2, terrain slope coefficient): heteroskedasticity-robust (HC3) β = 3.49 (95% CI, −0.19–7.17; $p = 0.063$); nonparametric bootstrap 95% CI (prefecture resampling, $B = 5{,}000$) = −0.03–7.09.

Table 4. Linear Regression Results for All Fracture Outcomes: Model 2 (Multivariable, N = 47)

Outcome	Slope β (95% CI)	*p*	Aging rate β (95% CI)	*p*	R^2	Adj. R^2
Hip (femur) fracture	**3.49 (0.11, 6.86)**	**0.043**	**5.53 (1.32, 9.74)**	**0.011**	**0.385**	**0.326**
Total fracture	4.47 (−0.88, 9.82)	0.099	**7.19 (0.51, 13.86)**	**0.035**	0.276	0.206
Humerus fracture	0.23 (−0.50, 0.96)	0.524	0.54 (−0.38, 1.46)	0.245	0.107	0.022
Forearm fracture	0.67 (−0.95, 2.30)	0.402	1.28 (−0.77, 3.33)	0.218	0.062	−0.027

All models adjusted for aging rate, fast walking rate, and population density. β, unstandardized regression coefficient; CI, confidence interval. **Bold** indicates $p < 0.05$.

For the hip (femur) outcome, heteroskedasticity-robust (HC3) inference and bootstrap CIs for the slope coefficient were borderline (Table 3 footnote).

Figure Legends

Figure 1. Scatter plot showing the association between habitable-area-weighted terrain slope (degrees) and hip fracture surgery rate (per 100,000 population) across 47 Japanese prefectures. Each point represents one prefecture. The solid line represents the simple linear regression fit.

Figure 2. Correlation heatmap of key study variables across 47 Japanese prefectures. Pearson correlation coefficients are displayed in each cell. Color intensity represents the magnitude of correlation (red = positive, blue = negative).

Figure 3. Prefectural distribution of habitable-area-weighted terrain slope (degrees) across Japan. Darker shading indicates steeper terrain. Prefectures in the Chugoku, Shikoku, and Kyushu mountain ranges show the highest slope values. **Note:** map lines delineate study areas and do not necessarily depict accepted national boundaries (here, prefectures used as the analytic geography).

Figure 4. Prefectural distribution of hip fracture surgery rate (per 100,000 population) across Japan. Darker shading indicates higher rates. Prefectures in mountainous regions tend to show elevated rates. **Note:** map lines delineate study areas and do not necessarily depict accepted national boundaries (here, prefectures used as the analytic geography).

Figure 5. Bivariate choropleth map displaying the joint distribution of terrain slope (horizontal axis of legend) and hip fracture surgery rate (vertical axis of legend) by prefecture. The 3×3 color grid classifies each prefecture into one of nine categories based on tertile rankings of each variable. Purple-brown colors indicate high slope and high fracture rate; light grey indicates low on both dimensions. **Note:** map lines delineate study areas and do not necessarily depict accepted national boundaries (here, prefectures used as the analytic geography).

Figure 1

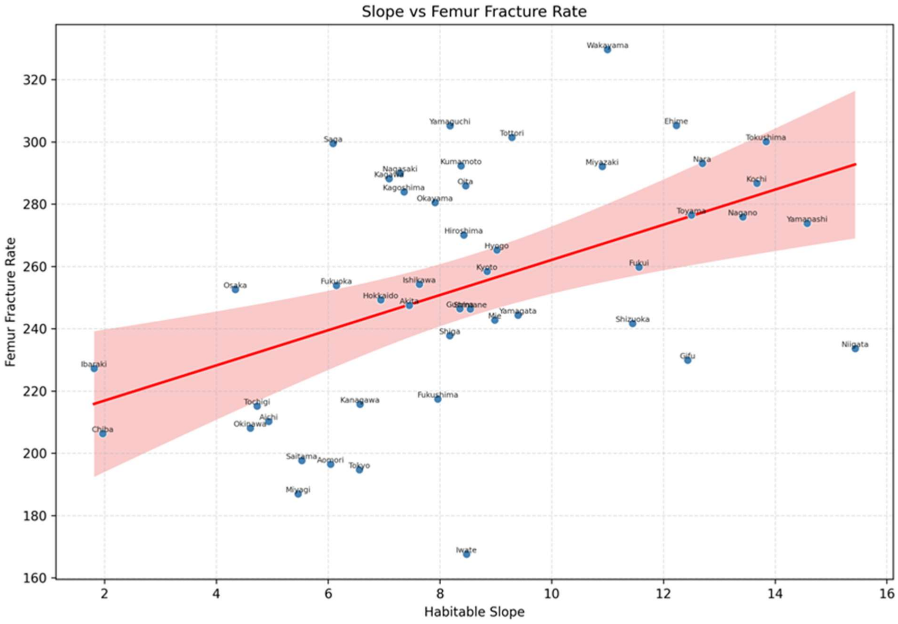
Slope vs Femur Fracture Rate
Femur Fracture Rate
Habitable Slope
320
300
280
260
240
220
200
180
160
2
4
6
8
10
12
14
16
Wakayama
Yamaguchi
Ehime
Tottori
Saga
Tokushima
Kumamoto
Miyazaki
Nara
Nagasaki
Kochi
Oita
Kagoshima
Okayama
Toyama
Nagano
Yamanashi
Hiroshima
Hyogo
Fukui
Kyoto
Osaka
Fukuoka
Ishikawa
Hokkaido
Akita
Yamagata
Shizuoka
Shiga
Niigata
Gifu
Ibaraki
Fukushima
Kanagawa
Tochigi
Aichi
Okinawa
Chiba
Saitama
Aomori
Tokyo
Miyagi
Iwate

Figure 2

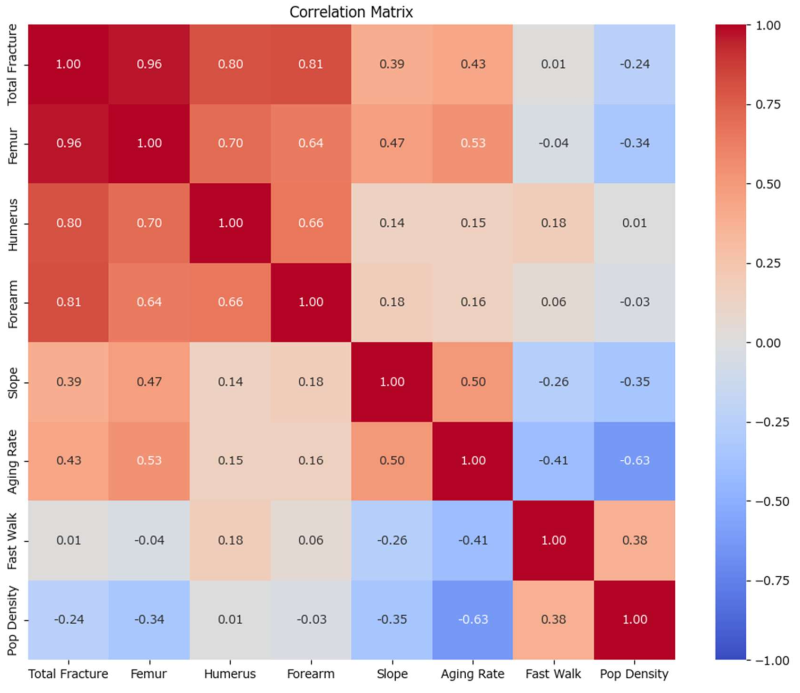
Correlation Matrix
Total Fracture
Femur
Humerus
Forearm
Slope
Aging Rate
Fast Walk
Pop Density
1.00
0.96
0.80
0.81
0.39
0.43
0.01
-0.24
0.70
0.64
0.47
0.53
-0.04
-0.34
0.66
0.14
0.15
0.18
0.16
0.06
-0.03
0.50
-0.26
-0.35
-0.41
-0.63
0.38
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

Figure 3

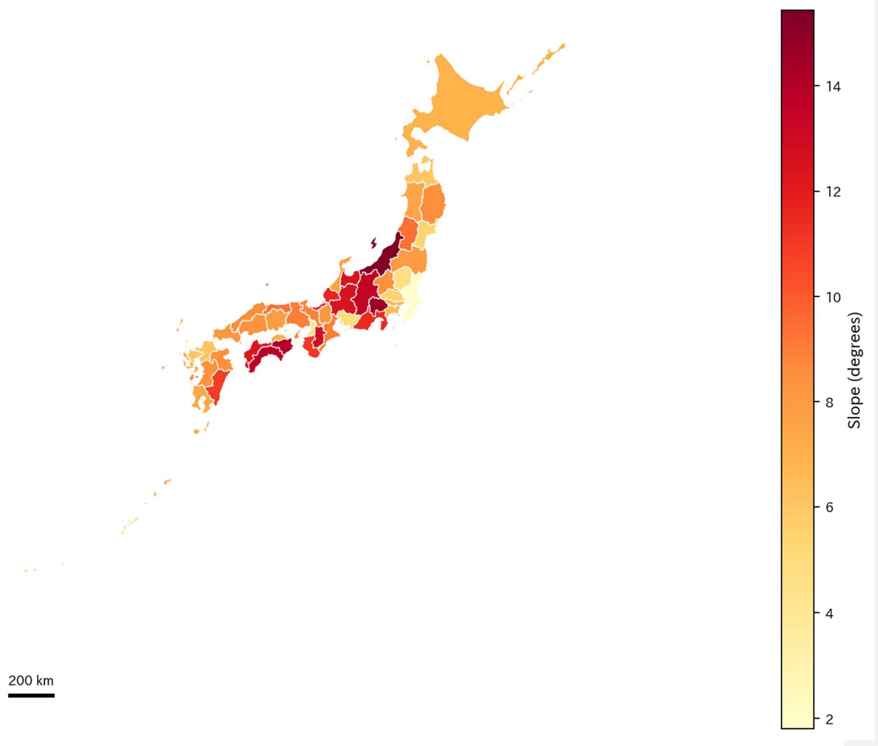

Figure 4

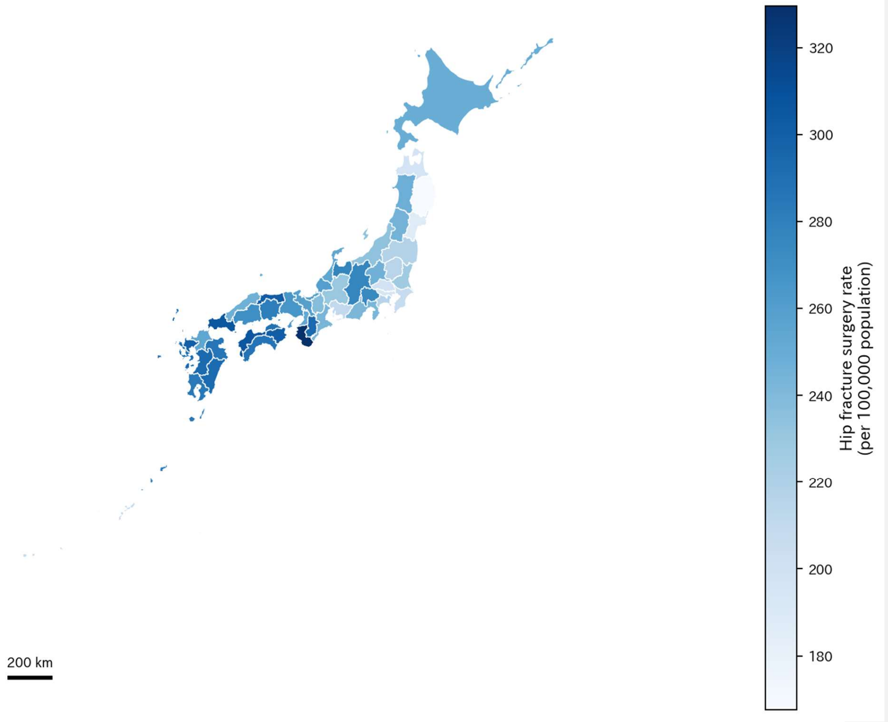
200 km
320
300
280
260
240
220
200
180
Hip fracture surgery rate
(per 100,000 population)

Figure 5

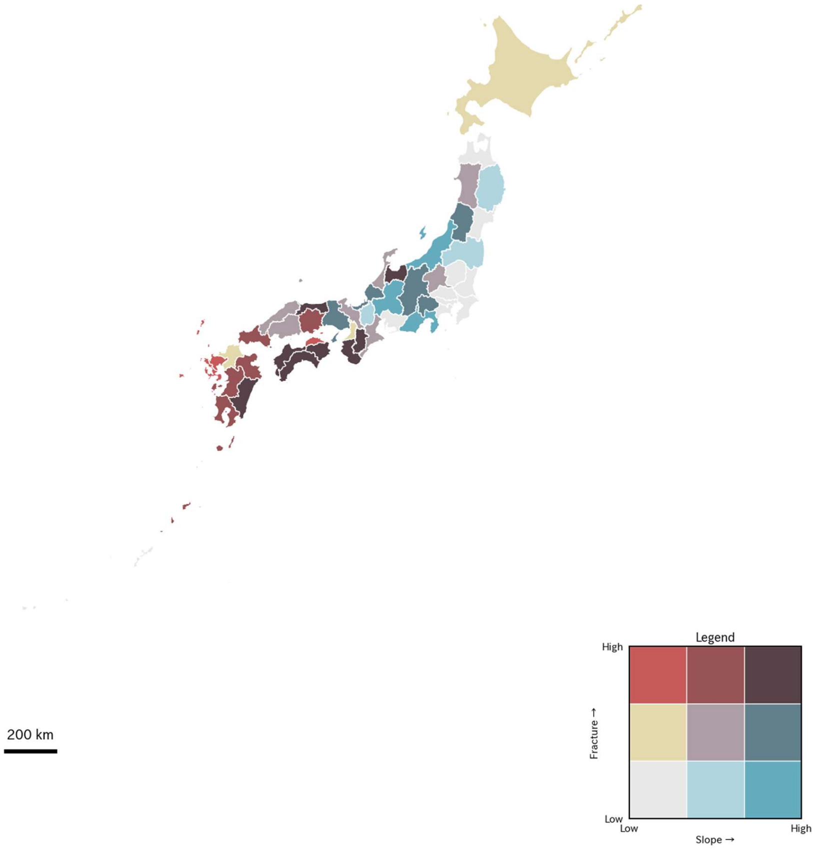

Figure 5. Bivariate Distribution of Terrain Slope
and Hip Fracture Surgery Rate by Prefecture